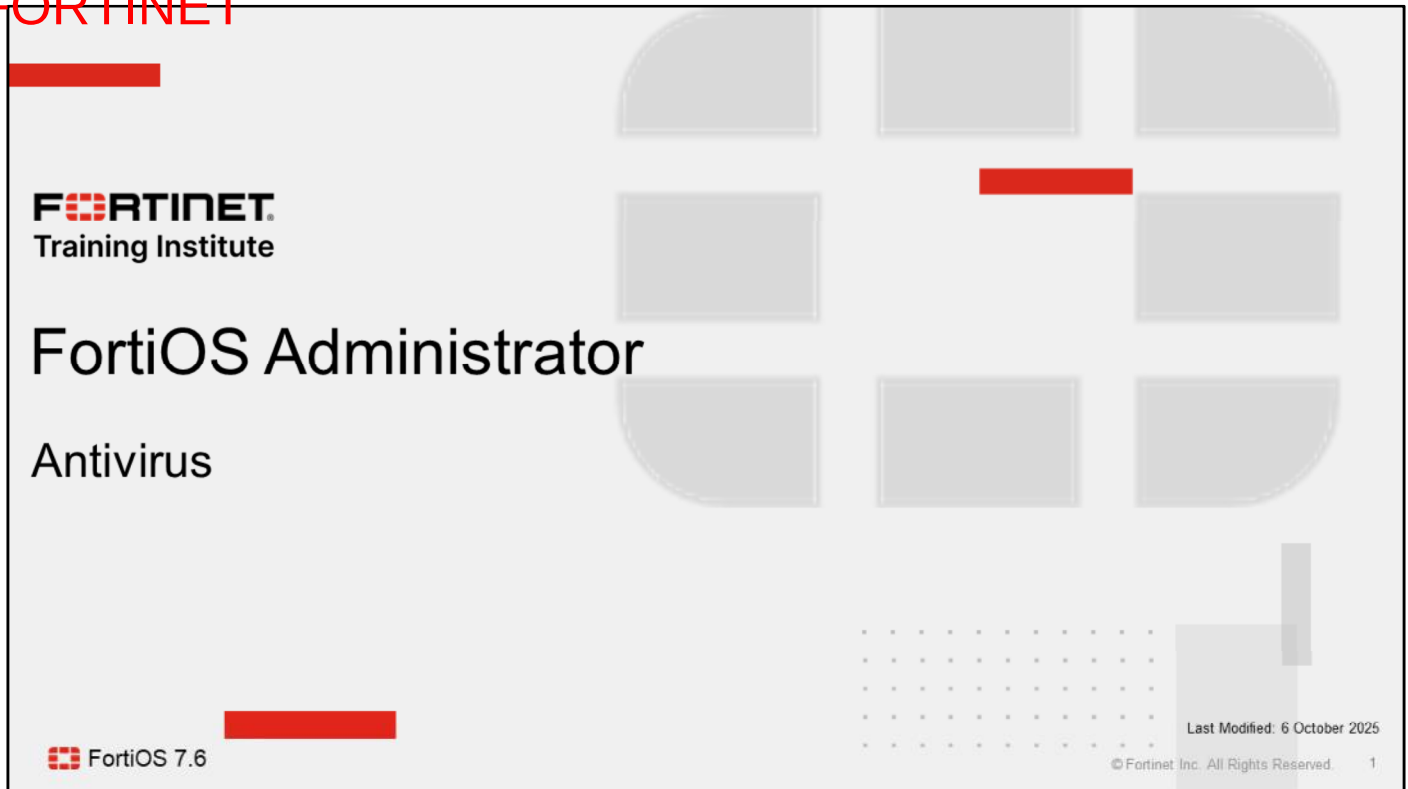


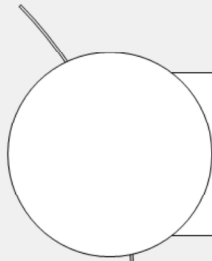
DO NOT REPRINT
© FORTINET



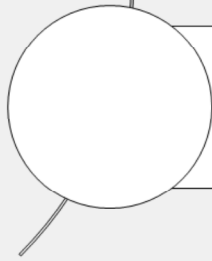
In this lesson, you will learn how to use FortiGate to protect your network against malware.

DO NOT REPRINT
© FORTINET

Lesson Overview



Antivirus Scanning Modes



Troubleshooting



© Fortinet Inc. All Rights Reserved. 2

In this lesson, you will learn about the topics shown on this slide.

DO NOT REPRINT
© FORTINET

Antivirus Scanning Modes

Objectives

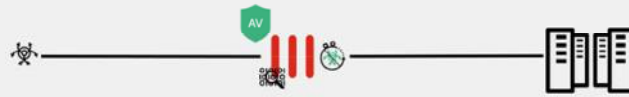
- Configure the antivirus profile in flow-based inspection mode
- Configure the antivirus profile in proxy-based inspection mode
- Configure protocol options

After completing this section, you should be able to achieve the objectives shown on this slide.

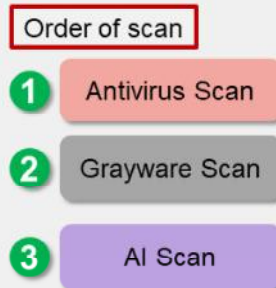
By demonstrating competence in antivirus basics, you will be able to understand and apply antivirus on FortiGate.

DO NOT REPRINT
© FORTINET

Antivirus Techniques



- Antivirus scanning engine uses antivirus signature databases to identify malicious codes
- Signature databases are updated in real time by FortiGuard antivirus service
- Antivirus scanning techniques:
 - Antivirus scan
 - Grayware scan
 - AI scan



© Fortinet Inc. All Rights Reserved. 4

A primary function of an antivirus scan is to detect and stop viruses that could cause harm to your system or compromise the security of your connected devices. It can be installed on individual endpoints (FortiClient), or it can operate as an antivirus engine to perform traffic inspection inside a next generation firewall (NGFW). The FortiGate antivirus engine operates by leveraging the information stored in signature databases that are updated by the FortiGuard antivirus service in real time. These signature databases are essentially vast repositories that contain detailed profiles of known and previously unknown viruses.

FortiGate uses many techniques to detect viruses. These detection techniques include:

- Antivirus scan: This scan is the first, fastest, simplest way to detect malware. It detects viruses that are an exact match for a signature in the antivirus database.
- Grayware scan: This scan detects unsolicited programs, known as grayware, that have been installed without the user's knowledge or consent. Often, grayware can be detected with a simple FortiGuard grayware signature.
- Artificial intelligence (AI) scan: The AI-based scan engine integrates with the regular antivirus scanning to detect zero-day malware in Windows portable executable (PE) files. Zero-day malware are new, unknown malware that have not been investigated by antimalware vendors and therefore have no existing associated signature that can detect them. FortiGuard Labs uses many malware samples to train the AI scanner to identify file features that comprise zero-day malware. Files detected by the AI scanner are identified with the W32/AI.Pallas.Suspicious virus signature.

If all antivirus features are enabled, FortiGate applies the following scanning order: antivirus scan, followed by grayware scan, followed by AI scan.

Antivirus Techniques (Contd)

- FortiGate uses several industry-standard techniques for antivirus protection

- Signature-based detection
- Virus outbreak prevention
- External malware block list
- EMS threat feed

Select the techniques for antivirus protection

Security Profiles > AntiVirus

Edit AntiVirus Profile

Name: default

Comments: Scan files and block viruses. 29/255

AntiVirus scan: ☒ Block ☐ Monitor

Feature set: ☒ Flow-based ☐ Proxy-based

Virus Outbreak Prevention

Use FortiGuard outbreak prevention database: ☒ Block ☐ Monitor

Use external malware block list: ☐

Use EMS threat feed: ☐



FortiGate uses a variety of industry-standard techniques for antivirus protection. You can configure these techniques on the GUI and CLI. The most common techniques are:

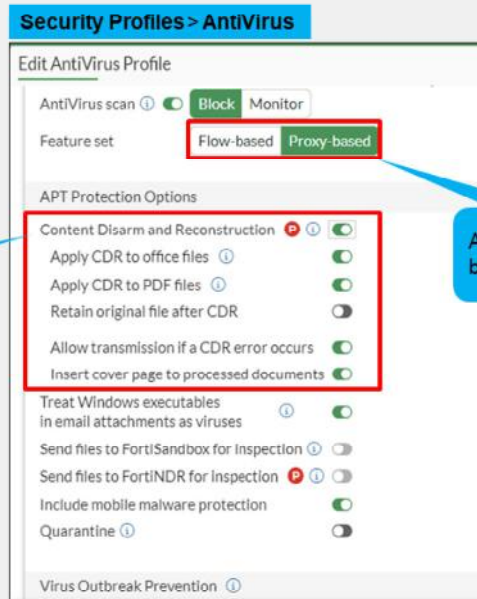
- Signature-based detection: Antivirus scan detects and compares the malicious file against virus signatures database. The FortiGuard antivirus service uses content pattern recognition language (CPRL), which is more efficient and accurate than traditional signature-based detection methods.
- Virus outbreak prevention: FortiGuard Virus Outbreak Protection Service (VOS) allows the FortiGate antivirus database to be subsidized with third-party malware hash signatures curated by FortiGuard. The hash signatures are obtained from FortiGuard's Global Threat Intelligence database. The antivirus database queries FortiGuard with the hash of a scanned file. If FortiGuard returns a match, the scanned file is deemed to be malicious. You do not have to enable to the antivirus engine scan to use this feature, but you must register FortiGate with a valid FortiGuard outbreak prevention license.
- External malware block list: FortiGate can enhance the antivirus database by linking dynamic external malware block lists to FortiGate. The lists are hosted on web servers and are available through HTTP/HTTPS URLs. To use this feature, you must create a new malware hash external connection under the Security Fabric. Then, you must select the external connector in the antivirus security profile. The hash lists hosted on these web servers can be MD5, SHA1, and SHA256 hashes, and are written on separate lines in a plaintext file.
- EMS threat feed: FortiGate receives malware feeds from FortiClient EMS, which itself gathers detected malware hashes from FortiClients. If an external malware blocklist and the FortiGuard outbreak prevention database are also enabled in the antivirus profile, the checking order is: antivirus local database, EMS threat feed, external malware blocklist, and FortiGuard outbreak prevention database. If the EMS threat feed and external malware blocklist contain the same hash value, then the EMS infection will be reported if both of them are blocked.

DO NOT REPRINT
© FORTINET

Antivirus Techniques (Contd)

- Content disarm and reconstruction (CDR)
- Behavior-based detection
- CIFS scanning
- AI/ML, behavioral, and human analysis

Enabling CDR
antivirus technique



Available with proxy-
based feature sets

- CDR allows FortiGate to sanitize Microsoft Office documents and PDF files (including those that are in ZIP archives) by removing active content, such as hyperlinks, embedded media, JavaScript, macros, and so on from the files without affecting the integrity of the textual content. When the client tries to download the file, FortiGate removes all exploitable content in real time and sends the disarmed file to the client. CDR is supported on HTTP, SMTP, POP3, and IMAP. CDR does not support flow-based inspection modes.
- Behavior-based detection: Submit suspected malicious files to FortiSandbox for inspection.
- CIFS scanning: File filtering and antivirus scanning on common internet file system (CIFS) traffic is supported.
- AI/ML, behavioral, and human analysis: These types of analysis help you identify, classify, and respond to threats. The antivirus engine AI malware detection model integrates into regular antivirus scanning to help detect potentially malicious files, such as malicious Windows PE or executable and linkable format (ELF) files, in order to mitigate zero-day attacks. Previously, this type of detection was handled by heuristics that analyzed file behavior. The antivirus engine AI package can be downloaded by FortiOS through FortiGuard, on devices with an active antivirus subscription. The ML detection setting is enabled by default at a per-VDOM level. Files detected by the antivirus engine AI are identified with the W32/AI.Pallas.Suspicious virus signature.

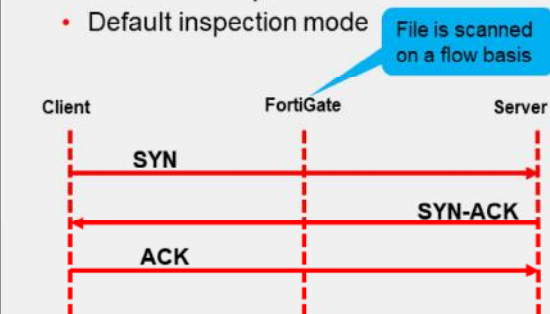
DO NOT REPRINT
© FORTINET

Inspection Modes

- Available inspection modes

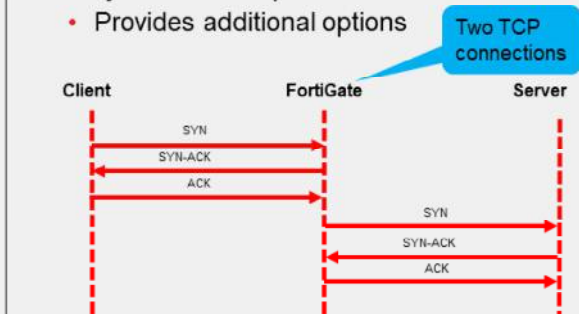
- Flow-based inspection

- Default inspection mode



- Proxy-based inspection

- Provides additional options



FortiGate uses different inspection modes for performance improvement and granularity.

Antivirus can operate in flow-based or proxy-based inspection mode.

Flow-based inspection takes a snapshot of content packets and uses pattern matching to identify security threats in the content. Proxy-based inspection reconstructs content that passes through FortiGate and inspects the content for security threats.

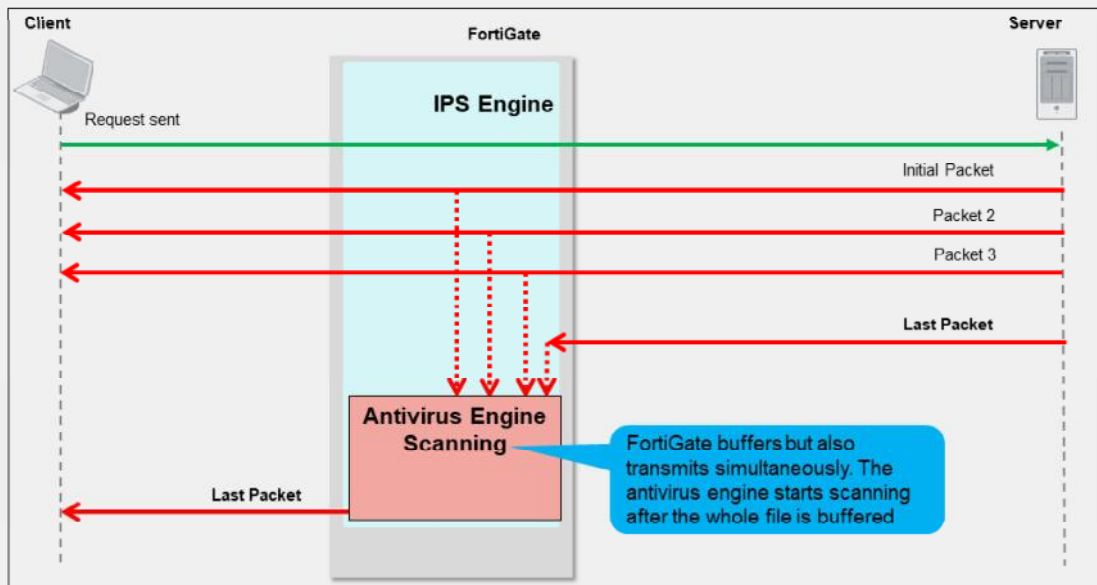
While both modes offer significant security, proxy-based mode provides more feature configuration options, while flow-based mode is designed to optimize performance.

Regardless of which mode you use, both use the full antivirus database (extended or extreme—depending on the CLI command `use-extreme-db` and the FortiGate model) and the scan techniques give similar detection rates. How can you choose which inspection mode to use?

If security is your priority, proxy-based inspection mode—with client comforting disabled—is more appropriate. If performance is your top priority, then flow-based inspection mode is more appropriate. Depending on the FortiGate model, flow-based pattern matching can be offloaded to CP8 or CP9 processors, and FortiGate models that support Nturbo can accelerate antivirus processing to enhance performance. NTurbo creates a special data path to redirect traffic from the ingress interface to the IPS engine, and from the IPS engine to the egress interface. This acceleration does not apply to proxy-based inspection.

DO NOT REPRINT
© FORTINET

Flow-Based Inspection Mode Packet Flow



© Fortinet Inc. All Rights Reserved. 8

Flow-based inspection is the default inspection mode for all new antivirus profiles.

Flow-based inspection mode uses a hybrid of two scanning modes: the default scanning mode and the legacy scanning mode. The scan method is determined by the intrusion prevention system (IPS) engine algorithm that is based on the type of file being scanned. The default antivirus scan automatically uses stream-based scanning in flow mode for HTML and JavaScript files. It triggers the legacy scan for unsupported configurations and file types.

This slide shows that the client sends a request and starts receiving packets immediately, but FortiGate also caches those packets at the same time. When the last packet arrives, FortiGate caches it and puts it on hold. Then, the IPS engine extracts the payload of the last packet, assembles the whole file, and sends it to the antivirus engine for scanning. If the antivirus scan does not detect any viruses, and the result comes back clean, the last cached packet is regenerated and delivered to the client. However, if the engine does find a virus, FortiGate resets the connection and does not send the last piece of the file. Although the receiver gets most of the file content, the file is truncated and therefore, can't be opened. The IPS engine also caches the URL of the infected file, so that if a second attempt to transmit the file is made, the IPS engine sends a block replacement message to the client instead of scanning the file again.

Because the file is transmitted at the same time, flow-based mode consumes more CPU cycles than proxy-based mode. However, depending on the FortiGate model, some operations can be offloaded to secure processing units (SPU) to improve performance.

DO NOT REPRINT
© FORTINET

Flow-Based Inspection Mode

- Default mode

Security Profiles > AntiVirus

Edit AntiVirus Profile

Name: default

Comments: Scan files and block viruses. 29/255

AntiVirus scan: ☒ Block ☐ Monitor

Feature set: ☒ Flow-based ☐ Proxy-based

Inspected Protocols:

- HTTP ☒
- SMTP ☒
- POP3 ☒
- IMAP ☒
- FTP ☒
- CIFS ☐

Action applied to the infected files

Only display profile with flow-based feature sets

Select protocols to be scanned

Enable AntiVirus profile in the firewall policy

Policy & Objects > Firewall Policy

Create New Policy

Name:

Incoming Interface:

Outgoing Interface:

Source:

Destination:

Schedule: ☒ always

Service:

Action: ☒ ACCEPT ☐ DENY

Inspection Mode: ☒ Flow-based ☐ Proxy-based

Firewall/Network Options

NAT: ☒

IP Pool Configuration: ☒ Use Outgoing Interface Address ☐ Use Dynamic IP Pool

Preserve Source Port: ☐

Protocol Options: ☒ reset default

Security Profiles

AntiVirus: ☒ default

Web Filter: ☐ Qsearch

Video Filter: ☐ default

DNS Filter: ☐ with-default

Application Control: ☐

IPS: ☐

File Filter: ☐

SSL Inspection: ☐ no-inspection

Default Inspection mode is Flow-based

Select the antivirus profile



Flow-based inspection mode is the default mode, and its configuration consists of two steps:

- Creating an antivirus profile that includes the inspected protocols and the action FortiGate will take when detects a virus-infected file.
- Applying the flow-based antivirus profile to a firewall policy.

DO NOT REPRINT
© FORTINET

Stream-Based Antivirus Scanning In Flow-Based Inspection

- Scanning for HTML and JavaScript files with antivirus engine 7.0
- Eliminates the need to cache entire file
- Improves memory usage

FortiGate will not use stream-based scanning if any of the following antivirus scanning configurations or features are enabled:

- ML-based malware detection
- Extreme antivirus database
- Greyware scan
- Mobile malware database
- External block list
- EMS threat feed
- FortiGuard outbreak prevention
- DLP
- File filter

Flow-based inspection mode

- Pattern matching can be offloaded to CP8 or CP9
- Priority on traffic throughput

Servers providing reliable service for large numbers of concurrent users



© Fortinet Inc. All Rights Reserved. 10

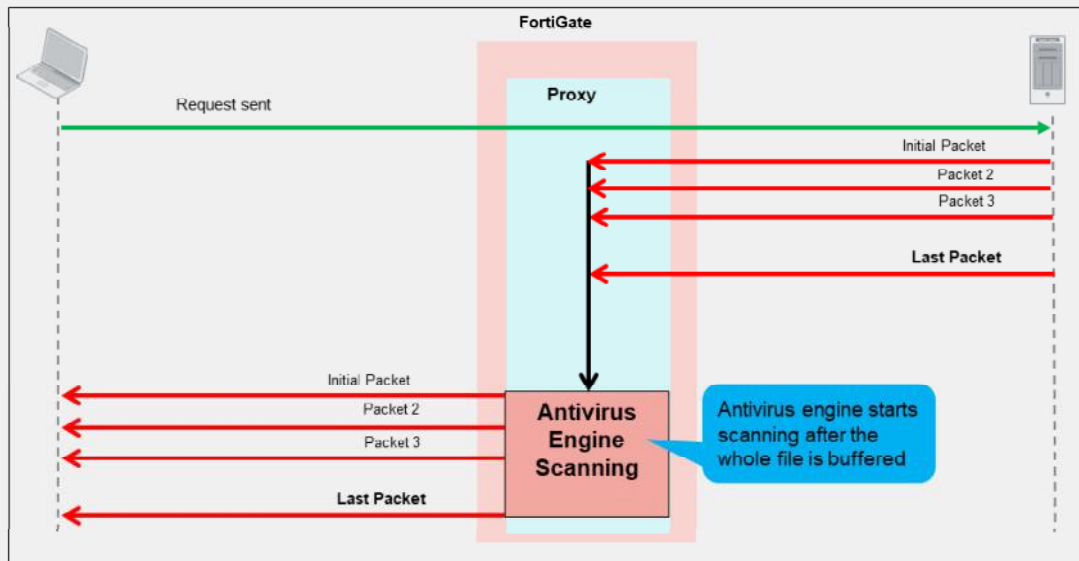
With antivirus engine 7.0, FortiGate uses stream-based antivirus scanning as the default scanning method for HTML or Javascript files. The antivirus engine determines the necessary amount of the file payload to buffer. Using the flow-based signatures, the antivirus engine scans the partial buffer in certain instances, eliminating the need to cache the entire file, which improves the memory usage.

FortiGate uses legacy antivirus scan for all file types that are not supported by the default antivirus scanning. The legacy antivirus scan is also used automatically when any of the following antivirus scanning features or configurations are enabled:

- ML-based malware detection (`set machine-learning-detection`) (enabled by default)
- Extreme antivirus database (`set use-extreme-db`)
- Antivirus potentially unwanted programs (PUP) and potentially unwanted applications (PUA) grayware checks
- Mobile malware database (`set mobile-malware-db`)
- External block list (`set external-blocklist`)
- EMS threat feed
- FortiGuard outbreak prevention
- DLP
- File filter

DO NOT REPRINT
© FORTINET

Proxy Inspection Mode Packet Flow



© Fortinet Inc. All Rights Reserved. 11

Proxy inspection packet flow steps:

1. The client sends a request to the server, for example, a file download.
2. FortiGate receives the packet and identifies it for proxy inspection. Then, the connection is split into two parts—client-to-FortiGate and FortiGate-to-server.
3. FortiGate receives and buffers the data from the server and starts the antivirus scan once the entire file is buffered. The FortiGate antivirus engine scans for malware in files, using its signature database.
4. If the engine does not detect a threat, the file is forwarded to the client. If the engine finds a virus, no packets are delivered to the end client and the proxy sends the replacement block message to the end client.

You can configure client comforting for HTTP and FTP from the `config firewall profile-protocol-options` command tree. This allows the proxy to slowly transmit data until it can complete the buffer and finish the scan. This prevents a connection or session timeout. No block replacement message appears in the first attempt because FortiGate is transmitting the packets to the end client.

DO NOT REPRINT
© FORTINET

Proxy Inspection Mode Enabled

- Configure the antivirus profile
 - **Feature set is proxy based**
- Provides additional antivirus support
 - MAPI and SSH protocol inspection
 - CDR
 - FortiNDR inspection
- Requires more than 2 GB RAM

Display profiles with
flow-based or proxy-
based feature sets

Available only in proxy-
based inspection mode

Security Profiles > AntiVirus

Edit AntiVirus Profile

Name: default
Comments: Scan files and block viruses. 29/255

AntiVirus scan: ☒ Block ☐ Monitor

Feature set: ☒ Flow-based ☒ Proxy-based

Inspected Protocols

HTTP ☒
SMTP ☒
POP3 ☒
IMAP ☒
FTP ☒
CIFS ☒
MAPI ☒
SSH ☒

APT Protection Options

Context Disarm and Reconstruction ☒
Treat Windows executables ☒
In email attachments as viruses ☒
Send files to FortiSandbox for inspection ☒
Send files to FortiNDR for inspection ☒
Include mobile malware protection ☒
Quarantine ☒



Proxy-based inspection mode is applied when you set **Feature set** to **Proxy-based**.

Unlike flow-based inspection mode, proxy-based inspection mode allows the profile to inspect the MAPI and SSH protocols traffic, as well as sanitize Microsoft documents and PDF files using the CDR feature.

FortiNDR can be used with antivirus profiles in proxy inspection mode (flow mode is currently not supported). FortiNDR inspects high-risk files and issues a verdict to the firewall based on how close the file features match those of malware.

To enhance performance and optimize memory usage, FortiOS no longer supports proxy-related features on FortiGate models with 2 GB RAM or less. Before you upgrade from a firmware version that supports proxy-related features to FortiOS 7.4.4 or a later version that no longer supports proxy-related features on FortiGate 2 GB RAM models, it is crucial that you carefully review the upgrade scenarios.

DO NOT REPRINT
© FORTINET

Firewall Policy With Proxy Inspection Mode

The screenshot shows the 'Policy & Objects > Firewall Policy' configuration page. The 'Create New Policy' form is displayed. The 'Inspection Mode' is set to 'Proxy-based'. The 'Security Profiles' section shows the following settings:

- Antivirus: default
- Web Filter: default
- Virus Filter: with-default
- DNS Filter: default
- Application Control: default
- IPS: default
- File Filter: default
- SSL Inspection: no-inspection

Available only in
proxy-based inspection mode

Set Inspection Mode
to Proxy-based

Proxy-based and flow-based
antivirus profiles available

The next step is to apply the proxy-based antivirus profile to a firewall policy. You must set the **Inspection Mode** to **Proxy-based** to be able to see proxy-based security profiles.

You can also view and select flow-based security profiles in a proxy-based policy rule. This is because other security features configured on the same policy rule may require the proxy, even if antivirus does not.

DO NOT REPRINT
© FORTINET

Stream-Based Antivirus Scanning In Proxy-Based Inspection

- Default scan mode in proxy mode
- Supports ZIP, GZIP, BZIP2, TAR, and ISO (ISO 9660) archive file types
- Supports HTTP(S), FTP(S), and SCP/SFTP protocols
- Inspects the contents of large archive files without buffering the entire file
- Decompresses and scans large archive files
- Identifies file types quickly
- Detects viruses effectively
- Can be disabled through the CLI

```
#config antivirus profile
edit <string>
  set feature-set proxy
  set scan-mode {default* | legacy}
next
end
```

legacy settings
disables stream-based
scanning

- Proxy-based inspection mode
 - Required for its additional options
 - Priority on network security

Protecting emails
received by mail servers
through SMTP or MAPI



© Fortinet Inc. All Rights Reserved. 14

Using proxy inspection antivirus allows you to use stream-based scanning, which is enabled by default. Stream-based scanning scans large archive files by decompressing the files and then scanning and extracting them at the same time. This process optimizes memory use to conserve resources on FortiGate. Viruses are detected even if they are in the middle or toward the end of these large files.

Stream-based scanning supports ZIP, GZIP, BZIP2, TAR, and ISO (ISO 9660) archive files types and HTTP(S), FTP(S), and SCP/SFTP protocols. It does not support HTTP POST. Stream-based scanning is not supported if the following features are enabled:

- DLP
- Quarantine
- FortiGuard outbreak prevention, external block list, and EMS threat feed
- Content disarm

DO NOT REPRINT © FORTINET

Antivirus Block Page

- Information available on the antivirus block page

The diagram illustrates the Antivirus Block Page and its components. The block page is titled "High Security Alert" and contains the following information:

- File name:** eicar.com
- Virus name:** EICAR_TEST_FILE
- Website host or URL:** http://10.200.1.254/eicar.com
- Quarantined File Name:** [disabled]
- Reference URL:** http://www.fortinet.com/ve?vn=EICAR_TEST_FILE

A red arrow points from the Reference URL to the FortiGuard Labs Threat Encyclopedia page, which is titled "Threat Encyclopedia" and shows details for the EICAR_TEST_FILE virus, including its ID (2172), release date (Oct 15, 1996), and description update date (Jun 05, 2015).

For antivirus scanning in proxy-based inspection mode (with client comforting disabled), the block replacement page is displayed *immediately* when a virus is detected.

For flow-based inspection mode scanning, if a virus is detected at the start of the stream, the block replacement page is displayed at the *first attempt*. If a virus is detected after a few packets have been transmitted, the block replacement page is *not* displayed. However, FortiGate caches the URL and can display the replacement page immediately, on the second attempt.

Note that if deep inspection is enabled, all HTTPS-based applications also display the block replacement message.

The block page includes the following:

- File name
- Virus name
- Website host and URL
- Username and group (if authentication is enabled)
- Link to FortiGuard Encyclopedia—which provides analysis, recommended actions (if any), and detection availability

You can go directly to the FortiGuard website to view information about other malware, and scan, submit, or do both, with a sample of suspected malware.

DO NOT REPRINT
© FORTINET

Configuring Protocol Options

- Available for both proxy-based and flow-based firewall policies

Policy & Objects > Firewall Policy

Policy & Objects > Protocol Options

Protocol	Action	Port
HTTP	Any	Specify 80
SMTP	Any	Specify 25
POP3	Any	Specify 110
IMAP	Any	Specify 143
FTP	Any	Specify 21,222,23
NNTP	Any	Specify 119
MAPI	Any	Specify 135
DNS	Any	Specify 53
CIFS	Any	Specify 445

Protocol options provide more granular control than antivirus profiles. You can configure protocol port mappings, common options, web options, email options, and more. Some options, like **Protocol Port Mapping**, apply only to proxy-based inspection.

When proxy-based antivirus scanning is enabled, FortiGate buffers files as they are downloaded. Once the entire file is captured, FortiGate begins scanning the file. The user must wait during the buffering and scanning procedure. After the scan is completed and if no infection is found, the file is sent to the next step in the process flow. If the file is large, this part of the process can take some time. In some cases, enough time that some users may get impatient and cancel the download.

The **Comfort Clients** option mitigates this potential issue by feeding a trickle of data while waiting for the scan to complete. The user is aware that processing is taking place and that there has not been a failure in the transmission. The slow transfer rate continues until the antivirus scan is complete. The transfer will proceed at full speed once the file is scanned successfully and does not contain any viruses.

Antivirus and other security profiles, such as web filtering, DNS filtering, and data loss prevention (DLP), use protocol options.

After configuring the protocol options, you should apply them to the firewall policy.

Protocol Options—Large Files

- By default, files that are bigger than the oversize limit are bypassed from scanning
- You can modify this behavior for all protocols

Common Options

Comfort Clients ☐

Block Oversized File/Email ☒

Threshold (MB) **10**

Applies to all protocols

Default value is 10 MB and the maximum value is hardware dependent

- You can enable logging of oversize files and adjust settings per protocol using the CLI

```
config firewall profile-protocol-options
edit <profile name>
  set oversize-log {enable|disable}
  config <protocol Name>
    set options oversize
    set oversize-limit <integer>
  end
end
end
```

Oversize file logging setting

Name of the specific protocol



So, what does the additional granularity provided by protocol options include? It allows you to block large files. You can also adjust the threshold for optimal performance in your network. The buffer limit varies by model. A smaller buffer minimizes proxy latency (for both scanning modes) and RAM usage, but that may allow viruses to pass through undetected. When a buffer is too large, clients may notice transmission timeouts. You must balance the two.

You can also disable the `oversize` option and adjust the `oversize-limit` per protocol from the `config firewall profile-protocol-options` command tree.

If you aren't sure about the value to set the `oversize-limit` to, you can temporarily enable the `oversize-log` to see if FortiGate is scanning large files frequently. You can then adjust the value accordingly.

DO NOT REPRINT
© FORTINET

Protocol Options—Compressed Files

- Archives are unpacked and files and archives within are scanned separately
- Password-protected archives cannot be decompressed
- Increasing the limits impacts memory usage

```
config firewall profile-protocol-options
edit <profile_name>
config <protocol_name>

set uncompressed-oversize-limit [1-<model_limit>]
set uncompressed-nest-limit [1-<model_limit>]

end
end
end
```

Oversize limit specific
to decompressed files

Nested archive limit



Large files are often compressed. When compressed files go through scanning, the compression acts like encryption: the signatures won't match. So, FortiGate must decompress the file in order to scan it.

Before decompressing a file, FortiGate must first identify the compression algorithm. Some archive types can be correctly identified using only the header. Also, FortiGate must check whether the file is password protected. If the archive is protected with a password, FortiGate can't decompress it and, therefore, can't scan it.

FortiGate decompresses files into RAM. Just like other large files, the RAM buffer has a maximum size. Increasing this limit may decrease performance, but it allows you to scan larger compressed files.

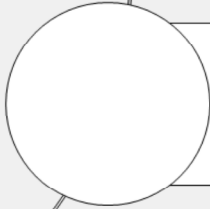
If an archive is nested—for example, if an attacker is trying to circumvent your scans by putting a ZIP file inside a ZIP file—FortiGate will try to undo all layers of compression. By default, FortiGate will attempt to decompress and scan up to 12-layers deep, but you can configure it to scan up to the maximum number supported by your device (usually 100). Increase this setting is not recommended because it increases RAM usage.

DO NOT REPRINT
© FORTINET

Lesson Progress



Antivirus Scanning Modes



Troubleshooting



© Fortinet Inc. All Rights Reserved. 19

Good job! You now understand antivirus scanning modes.

Now, you will learn about antivirus troubleshooting.

DO NOT REPRINT
© FORTINET

Troubleshooting

Objectives

- Monitor antivirus events
- Troubleshoot common antivirus issues



© Fortinet Inc. All Rights Reserved. 20

After completing this section, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in troubleshooting common antivirus issues, you will be able to configure and maintain an effective antivirus solution.

DO NOT REPRINT
© FORTINET

Antivirus Logs

Log & Report > Security Events

12 Events

AntiVirus ☒

Top Virus/Botnet	Action	Count
1132999808	Blocked	6
eicar.com	Blocked	6

12 events

Summary **Logs**

Date/Time: 2024-10-30 14:08:49 -> 2024-10-30 14:13:00

Search: [X] [Q] [AntiVirus] [Disk] [custom] [Details]

Date/Time	Service	Source	File Name	Virus/Botnet	User	Details	Action	Infection Type
2024/10/30 14:09:58	FTP	10.0.1.10	eicar.com	ECAR_TEST_FILE	Host: 10.200.1.254		Blocked	Malicious
2024/10/30 14:09:58	FTP	10.0.1.10	eicar.com	ECAR_TEST_FILE	Host: 10.200.1.254		Blocked	Malicious
2024/10/30 14:09:58	FTP	10.0.1.10	eicar.com	ECAR_TEST_FILE	Host: 10.200.1.254		Blocked	Malicious

Log entry when a virus is detected

Details on the virus with FortiGuard reference

Log Details

Application Control

Protocol: 6
Service: FTP

Data

File Name: eicar.com

Action

Action: Blocked
Threat: 2
Policy ID: Full_Access (1)
Policy UUID: b11ac58c-791b-51e7-4600-12f829a689d9
Policy Type: Firewall

Security

Level: Warning
Threat Level: Critical
Threat Score: 50

AntiVirus

Profile: default
Virus/Botnet: ECAR_TEST_FILE
Virus ID: 2,172
Reference: <https://fortiguard.com/eneyclopedia/virus/2172>
Detection Type: av-engine
Direction: incoming
Quarantine Skip: Quarantine-disabled
FortiSandbox Checksum: 275a021bbfb6489e54d471899f7db9d1663fcd95ec2fe2a2c4538aabb651fd0f
Submitted to FortiSandbox: false
Message: File is infected.

© Fortinet Inc. All Rights Reserved. 21

Logging is an important part of managing a secure network. When you enable unified threat management (UTM) logging in a firewall policy, you can find details on the **AntiVirus** log page under **Security Events**.

When the antivirus scan detects a virus, by default, it creates a log about which virus was detected, as well as the action, policy ID, antivirus profile name, and detection type. It also provides a link to more information on the FortiGuard website.

When you enable oversized file logging, a log entry is also created with the details that include the message, "Size limit is exceeded".

DO NOT REPRINT
© FORTINET

Forward Traffic Logs

Log & Report > Forward Traffic

[Search Icon] [Search] [Filter Icon] [Filter] [Sort Icon] [Sort] [Disk Icon] [Disk] [5 minutes -] [Details Icon] [Details]						
Date/Time	Source	Device	Destination	Application Name	Result	Policy ID
2024/10/30 14:26:53	10.0.1.10		34.107.221.62 (detectportal.firefox.com)	HTTP	✓ Accept (926 B / 1.17 kB)	Full Access (1)
2024/10/30 14:26:08	10.0.1.10		10.200.1.254	HTTP	⛔ Deny (Deny: UTM Blocked)	Full Access (1)
2024/10/30 14:25:56	10.0.1.10		10.200.1.254	HTTP	⛔ Deny (Deny: UTM Blocked)	Full Access (1)
2024/10/30 14:25:23	10.0.1.10		10.200.1.254	HTTP	⛔ Deny (Deny: UTM Blocked)	Full Access (1)
2024/10/30 14:22:09	10.0.1.200		96.45.43.45 (ns1.safeguard.net)	top/833	✓ Accept (16.75 kB / 17.25 kB)	Full Access (1)

Forward traffic log entry

Security log details

Log Details	
Details	Security
Antivirus	
Agent	Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/117.0
Submitted to FortiGuard	false
Direction	incoming
Destination UUID	7bc87d34-7916-51e7-3d5b-71812a61b99e
Detection Type	av-engine
Log event original timestamp	1730.312.755.004.399.400
Event Type	infected
File Name	eicar.com
HTTP request method	GET
Infection Type	Malicious
Level	Warning
Profile	default
Quarantine Skip	Quarantine-disabled
Reference	https://fortiguard.com/securitylogs/2172
Referrer URI	http://10.200.1.254/test_av.html
Source UUID	703e6ff6-791a-51e7-daa0-9859ce6c1d02
Sub Type	virus
Transaction ID	1
Type	utm
Timezone	-0700
URL	http://10.200.1.254/eicar.com
Virus/Botnet	EICAR_TEST_FILE
Virus Category	Virus
Virus ID	2.172
Details	URL: http://10.200.1.254/eicar.com

You can also view log details on the **Forward Traffic** page, where firewall policies record traffic activity. You can find a summary of the traffic on which FortiGate applied an antivirus action in the corresponding security details.

DO NOT REPRINT
© FORTINET

Security Dashboard

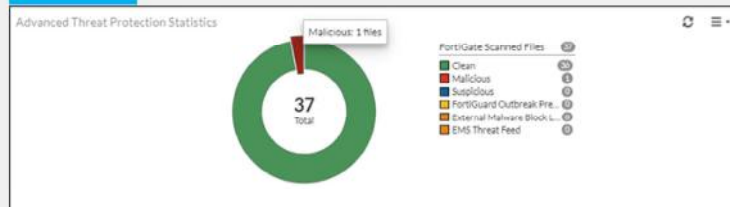
- Security widget and dashboard allow you to monitor your network

Dashboard > Security



Drill down for further details

Dashboard



Security widget

You can also use the **Security** dashboard to view relevant information regarding threats to your network. The security dashboard organizes information into source and destination, and allows you to drill down to see session logs details.

You can add the **Advanced Threat Protection Statistics** widget to the dashboard for monitoring purposes.

DO NOT REPRINT
© FORTINET

Troubleshooting Common Antivirus Issues

- Verify FortiGuard antivirus license

System > FortiGuard

FortiGuard Distribution Network

License Information	
Entitlement	Status
☒ Advanced Malware Protection	✔ Licensed (Expiration Date: 2026/01/17)
AI Malware Detection Model	⌚ Version 2.05360
AntiVirus Definitions	⌚ Version 90.01635
AntiVirus Engine	⌚ Version 7.00030
Mobile Malware	⌚ Version 90.01635
Outbreak Prevention	✔ Licensed (Expiration Date: 2026/01/17)

Upgrade Database

Valid license

- Force FortiGate to check for new antivirus updates

```
# execute update-av
```

- Run the real-time update debug to isolate update-related issues

```
# diagnose debug application update -l
# diagnose debug enable
# execute update-av
```



Viruses are constantly evolving and you must have the latest antivirus definitions version to ensure correct protection. With a valid license, FortiGate checks regularly for updates. If an antivirus profile is applied on at least one firewall policy, you can also force an update of the antivirus definitions database with the CLI command `execute update-av`.

If you are having issues with the antivirus license or FortiGuard updates, start troubleshooting with basic connectivity tests. Most of the time, issues related to updates are caused by connectivity problems with FortiGuard servers. You can do the following to handle common antivirus issues:

- Make sure that FortiGate has a stable internet connection and can resolve DNS (`update.fortinet.net`).
- If there is another firewall between FortiGate and the internet, make sure TCP port 443 is open and traffic is allowed from and to FortiGate.
- If you continue to see issues with the update, run the real-time debug command to identify the problem.

DO NOT REPRINT
© FORTINET

Troubleshooting Common Antivirus Issues (Contd)

- Unable to detect viruses even with a valid contract?

Check firewall policy configuration

In proxy-based inspection mode, verify the protocol port mapping

Verify the antivirus profile applied

For encrypted protocols, you must select deep inspection

The screenshot displays the configuration for a firewall policy named 'Full Access'. Key settings include:

- Name:** Full Access
- Incoming Interface:** port3
- Outgoing Interface:** port1
- Source:** LOCAL_SUBNET
- Destination:** all
- Schedule:** always
- Service:** ALL
- Action:** ACCEPT (checked), DENY
- Inspection Mode:** Flow-based, Proxy-based (checked)
- Firewall/Network Options:**
 - NAT:** Enabled
 - IP Pool Configuration:** Use Outgoing Interface Address, Use Dynamic IP Pool
 - Preserve Source Port:** Disabled
 - Protocol Options:** NAT default
- Security Profiles:**
 - AntiVirus:** AV default
 - Web Filter:** Disabled
 - Video Filter:** Disabled
 - DNS Filter:** Disabled
 - Application Control:** Disabled
 - IPS:** Disabled
 - File Filter:** Disabled
 - SSL Inspection:** deep-inspection



What if you have a valid contract and updated database, but you are still having issues catching viruses? Start troubleshooting for basic configuration errors. Most of the time, issues are caused by misconfiguration on the device. You can do the following to verify:

- Make sure that the correct antivirus profile is applied on the correct firewall policy.
- Make sure that the correct protocol port is configured when proxy-based inspection mode is used.
- Make sure that you are using the correct antivirus profile and SSL/SSH inspection on all firewall policies.

Troubleshooting Common Antivirus Issues (Contd)

- Check useful antivirus commands

```
# get system performance status
```

```
Virus caught: 100 total in 1 minute
```

Displays virus statistics for the last one minute

```
# diagnose antivirus database-info
```

```
version: 90.01635(04/22/0022 13:26)
```

```
atdb found 1 loaded 1
  virus ID count      29630
  grayware ID count   140
  signature ID count  49988
etdb found 1 loaded 1
  virus ID count      60712
  grayware ID count   4429
  signature ID count  806735
exdb found 1 loaded 0
  virus ID count      0
  grayware ID count   0
  signature ID count  0
```

Displays current antivirus database information

```
# diagnose autoupdate versions
Virus Definitions
```

Displays version information

```
-----
Version: 90.01635 signed
Contract Expiry Date: Mon Jan 19 2026
Last Updated using manual update on Mon
Apr 25 13:52:18 2022
Last Update Attempt: Wed Sep 13 06:27:50
2023
Result: No Updates
```

```
# diagnose antivirus test "get scantime"
antivirus test (manager)
```

```
0~5s: 0
5~10s: 0
10~15s: 0
15~20s: 0
20~25s: 0
25~30s: 0
>30s: 0
```

Displays scan times for infected files



To troubleshoot further common antivirus issues, you can check the information provided by the following commands:

- `get system performance status`: Displays statistics for the last one minute.
- `diagnose antivirus database-info`: Displays current antivirus database information.
- `diagnose autoupdate versions`: Displays current antivirus engine and signature versions.
- `diagnose antivirus test "get scantime"`: Displays scan times for infected files.

DO NOT REPRINT
© FORTINET

Knowledge Check

1. Which additional features of an antivirus profile are available in proxy-based inspection mode?
☒ A. MAPI, SSH, CDR, and FortiNDR
☐ B. Full and quick
2. What does the oversize files logging setting do?
☒ A. Enables logging of all files that exceed the oversize limit
☐ B. Logs all files that are over 5 MB
3. Which feature does not support stream-based scanning?
☒ A. Data loss prevention (DLP)
☐ B. Signature-based detection



DO NOT REPRINT
© FORTINET

Lesson Progress



Antivirus Scanning Modes



Troubleshooting



© Fortinet Inc. All Rights Reserved. 28

Congratulations! You have completed this lesson.

Now, you will review the objectives covered in this lesson.

DO NOT REPRINT
© FORTINET

Review

- ✓ Apply the antivirus profile in flow-based inspection modes
- ✓ Apply the antivirus profile in proxy-based inspection modes
- ✓ Compare inspection modes
- ✓ Configure protocol options
- ✓ Monitor antivirus events
- ✓ Troubleshoot common antivirus issues



This slide shows the objectives that you covered in this lesson.

By mastering the objectives covered in this lesson, you learned how to use FortiGate features and functions to protect your network against viruses.